
Example Namespace

NASA Planetary Data System

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The Balloon Observation Platform for Planetary Science (BOPPS) mission was an Earth-based stratospheric balloon with a gondola housing the refurbished Stratospheric Terahertz Observatory (STO) Telescope and infrared camera. It flew on 26 September 2014 and obtained images of asteroid (1) Ceres and comets C/2012 K1 (PanSTARRS), C/2013 A1 (Siding Spring), and C/214 E2 (Jacques).

INTRODUCTION

No documentation was provided with the original BOPPS dictionary. The minimal information here has been extracted from the source files.

OVERVIEW OF THE BOPPS LOCAL DATA DICTIONARY

The Balloon Observation Platform for Planetary Science (BOPPS) mission was an Earth-based stratospheric balloon with a gondola housing the refurbished Stratospheric Terahertz Observatory (STO) Telescope and infrared camera. It flew on 26 September 2014 and obtained images of asteroid (1) Ceres and comets C/2012 K1 (PanSTARRS), C/2013 A1 (Siding Spring), and C/214 E2 (Jacques).

ORGANIZATION OF CLASSES AND ATTRIBUTES

The BOPPS dictionary contains three classes which are added to the product labels individually as needed.

3.1 <bopps:Observation_Parameters>

This class contains attributes for observation timing, filter, pointing, and similar quantities.

3.2 <bopps:Telescope_Geometry>

This class provides geometry specific to the gondola-based observing environment, as well as some elements (like airmass) that were not available in discipline dictionaries at the time the labels were designed.

3.3 <bopps:Instrument_Temperature>

This class, which may occur more than once, provide temperatyre information recorded by a specified sensor.

BALLOON OBSERVATION PLATFORM FOR PLANETARY SCIENCE (BOPPS) NAMESPACE OUTLINE

Unlike most dictionaries, the BOPPS namespace does not have a wrapper class as a single entry point. The following classes may be invoked as needed in any order. The only class that can logically be repeated is the <bopps:Instrument_Temperature> class. In practice, when a class appears it typically contains all the attributes shown below, although most are optional.

Note that in the 1.x versions of this dictionary, the class names were all lowercase. This was changed in both the labels and in this dictionary as part of the migration of this dataset to the PDS cloud environment.

```
<bopps:Observation_Parameters>
  <bopps:spacecraft_clock_start_count>
  <bopps:spacecraft_clock_stop_count>
  <bopps:total_integration_time>
  <bopps:average_dn_for>
  <bopps:average_e_for>
  <bopps:max_average_dn_for>
  <bopps:min_average_dn_for>
  <bopps:max_average_e_for>
  <bopps:min_average_e_for>
  <bopps:filter>
  <bopps:filter_wavelength>
  <bopps:product_type>
  <bopps:observation_description>
  <bopps:pointing_descript>
  <bopps:centx>
  <bopps:centy>

<bopps:Telescope_Geometry>
  <bopps:gondola_latitude>
  <bopps:gondola_longitude>
  <bopps:gondola_altitude>
  <bopps:ra_cmd>
  <bopps:dec_cmd>
  <bopps:deviation_az>
  <bopps:deviation_el>
  <bopps:nod>
  <bopps:starfix>
  <bopps:solar_el>
  <bopps:boresight_el>
  <bopps:airmass>
```

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```
<bopps:Instrument_Temperature>  
  <bopps:measured_at>  
  <bopps:temperature>  
  <bopps:min_temperature>  
  <bopps:max_temperature>
```